Quantum Liquid Neural Networks for ODE / PDE Solving & Forecasting

A pre-registered, falsifiable benchmark against matched classical baselines

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Advisor update — July 2026

target venue: PRX Quantum

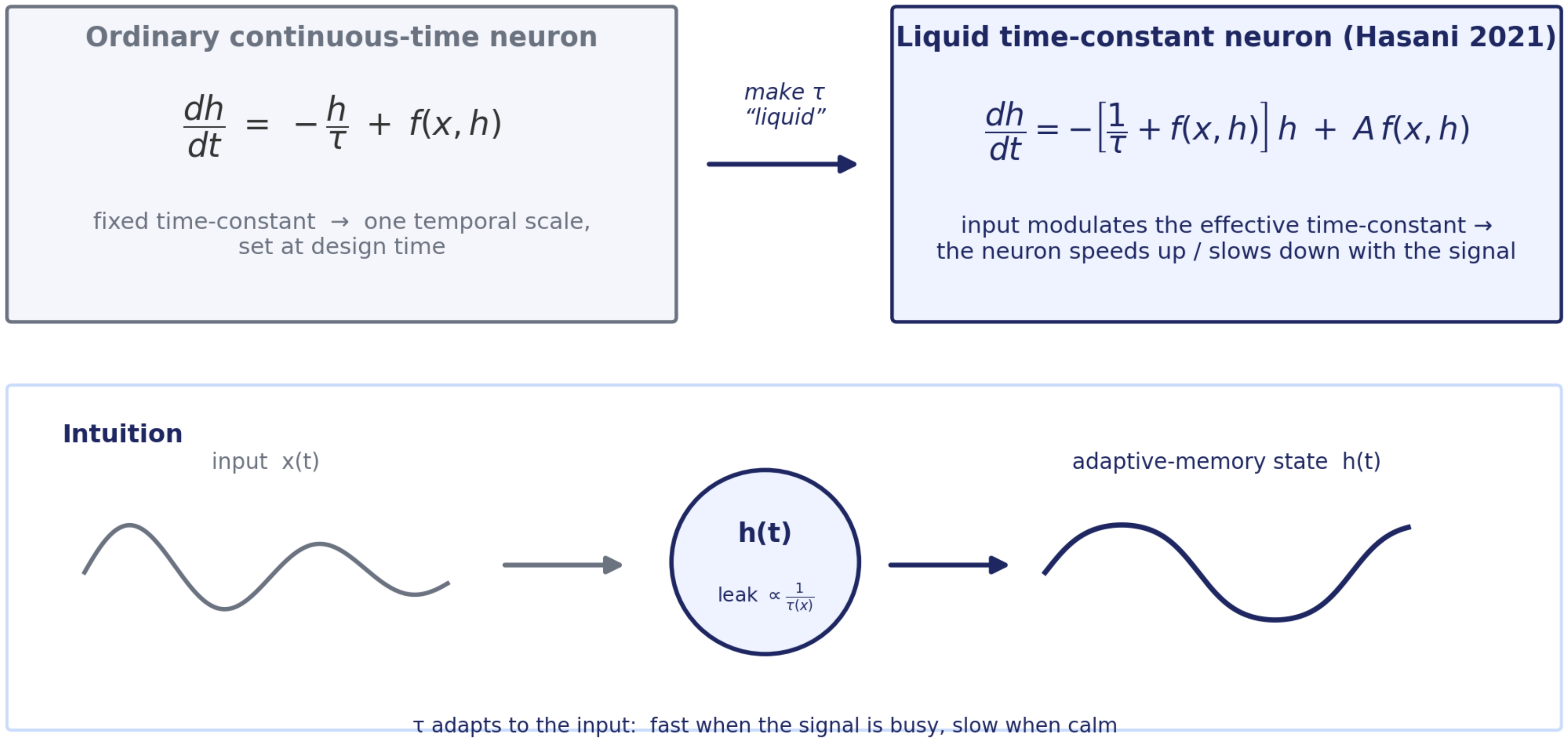
The research question

- **Do quantum PINNs hold a regime-dependent advantage — better on smooth/periodic dynamics than on broadband/chaotic dynamics?**
- Theory motivation: data-reuploading circuits are truncated Fourier series (Schuld 2021) → built-in bias toward smooth, low-frequency functions
- Field problem: most QML papers compare against under-tuned classical baselines (Bowles, Ahmed & Schuld 2024)
- **Our answer: pre-register the hypothesis, lock the decision rule, tune BOTH sides, then let the data decide**

9 quantum model families	**5** classical baselines
8 ODE / PDE systems	**405+** training cells, 0 errors
22 pre-registration amendments — every choice disclosed	

Bowles, Ahmed, Schuld (2024) "Better than classical? The subtle art of benchmarking QML models," arXiv:2403.07059

Background: what is a Liquid Neural Network?

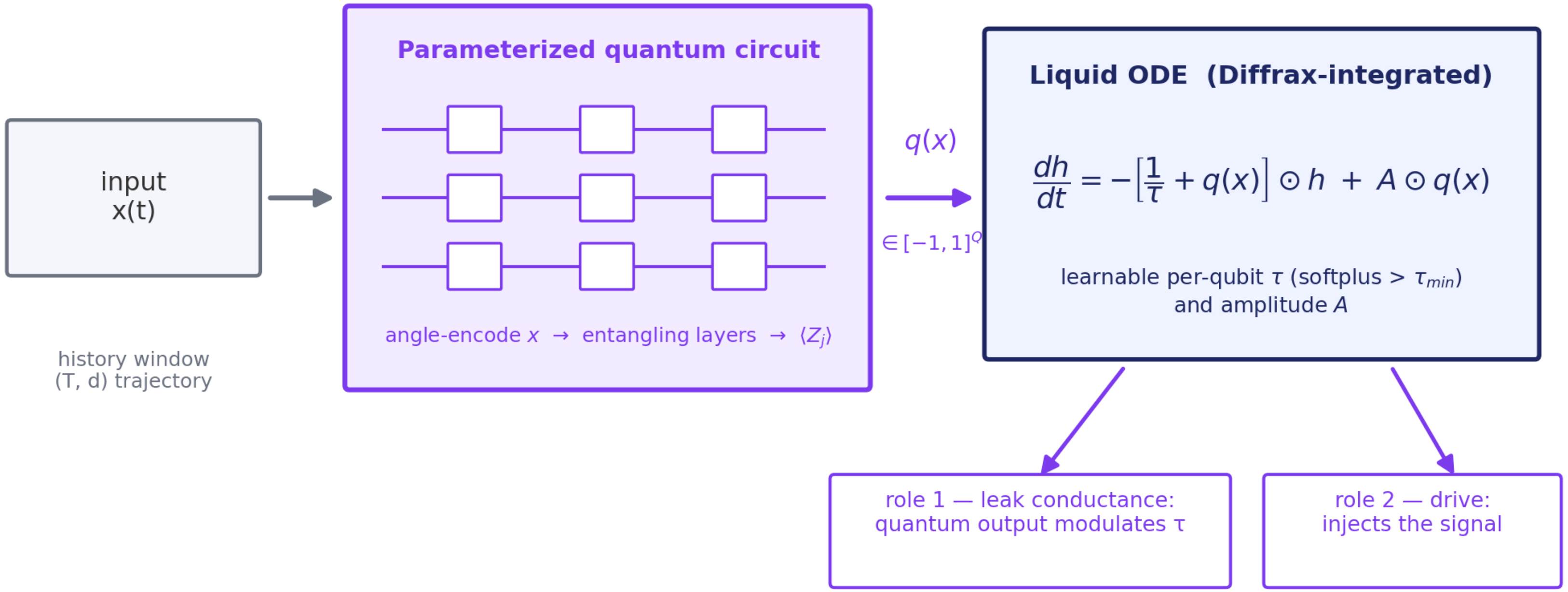


- **MIT (Hasani et al., AAAI 2021): neurons are tiny ODEs**
- The time-constant τ is modulated by the input — the network's memory adapts to the signal ("liquid")
- Inspired by C. elegans neurons (302 neurons, complex behavior)
- Strong on time-series; the basis of MIT's autonomous-driving NCPs

Diagram: ours, after Hasani et al. 2021 (arXiv:2006.04439). dh/dt = −[1/τ + f(x,h)]h + A·f(x,h)

Where the “quantum” goes in our model

Our cell: swap the classical nonlinearity f(x) for a quantum circuit

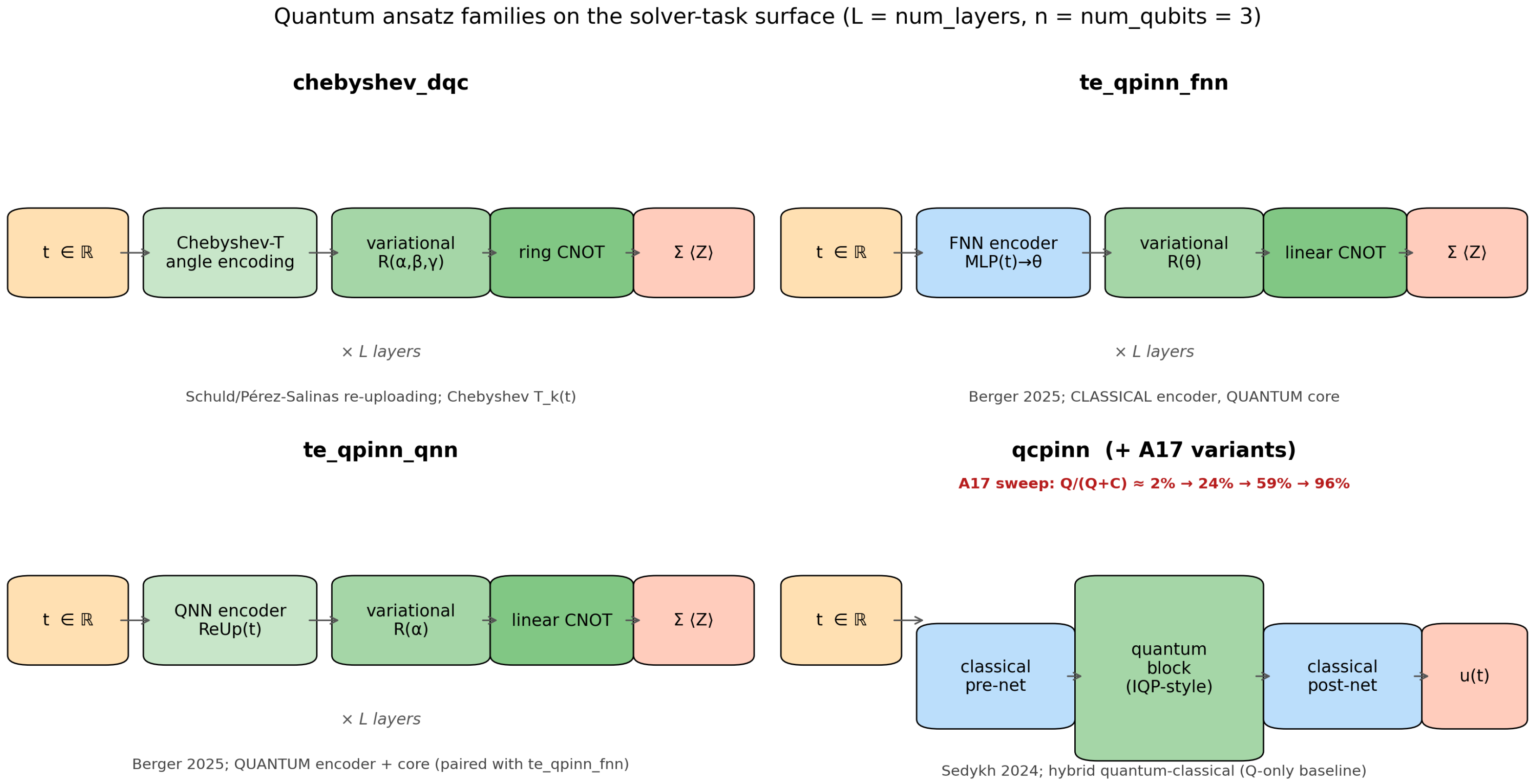


Fairness 2×2: remove τ-leak → non-liquid quantum cell | swap PQC for MLP → classical LTC | remove both → plain Neural-ODE

- **We replace the classical nonlinearity f(x) with a parameterized quantum circuit**
- PauliZ readout q(x) plays BOTH liquid roles: leak conductance and drive
- τ and amplitude A stay classical and learnable
- Every piece is separable → clean ablations (next slides)

Solver task uses the same circuits as PINN function approximators (no ODE cell; fairness is PINN-vs-PINN there).

The model zoo — every quantum family vs every baseline

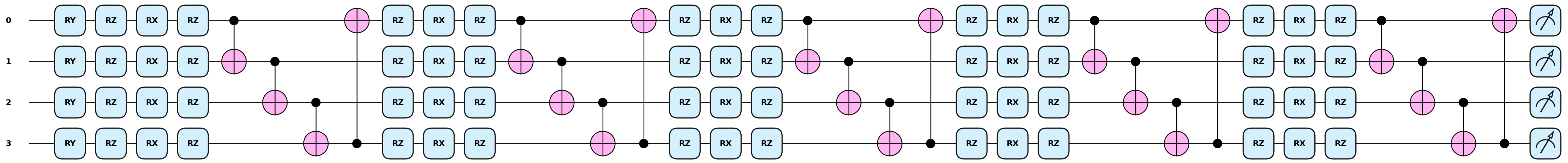


- Quantum (solver): chebyshev_dqc, te_qpinn_fnn, te_qpinn_qnn, qcpinn (+2D ports)
- Quantum (forecaster): data-reuploading, hardware-efficient, strongly-entangling, rf_qrc + non-liquid ablations
- Classical: PINN, Neural-ODE, classical LTC, plain MLP, known-structure skyline
- **All capacity-matched; all-to-all head-to-head in the paper**

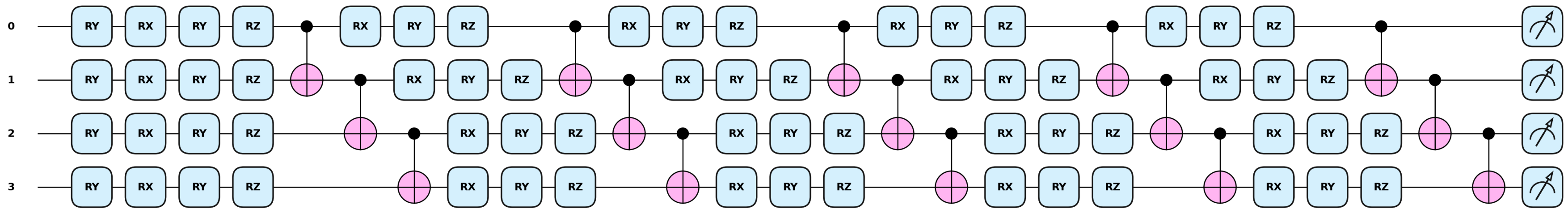
Every family implemented against its published spec; faithfulness manifest in refs/CIRCUIT_SPECS.md.

The solver circuits — printed from the code (PennyLane)

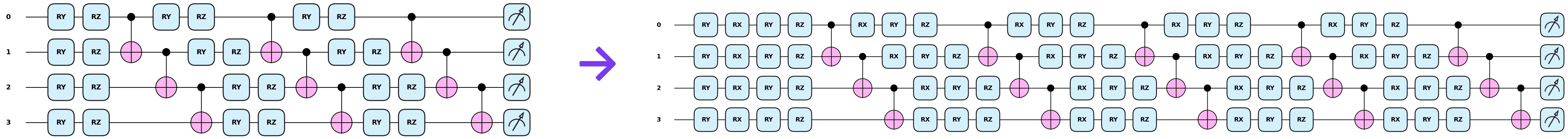
chebyshev_dqc — Chebyshev tower RY(2(j+1)·arccos x) → 5× [RZ·RX·RZ + ring CNOT] → ⟨Σ Zj⟩ (n=4, L=5)



te_qpinn_fnn — classical MLP ϕ(t) → RY(ϕk·t) embed → 5× HEA [RX·RY·RZ + CNOT chain] → ⟨Σ Zj⟩ (n=4, L=5)

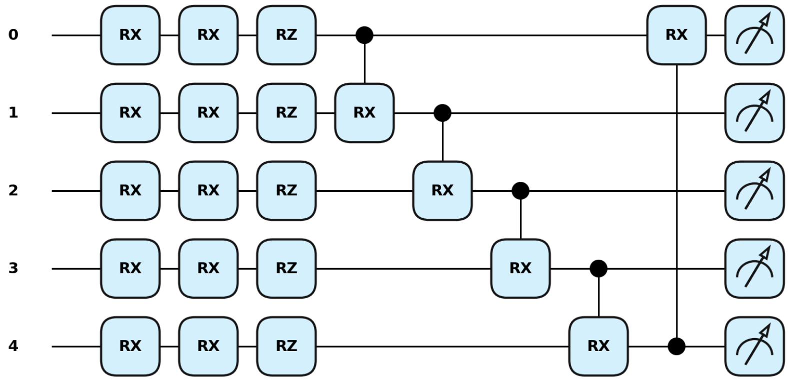


te_qpinn_qnn — stage 1: quantum embedding (K=3) → αk = π·⟨Zk⟩ … … stage 2: RY(αk) re-encode → 5× HEA → ⟨Σ Zj⟩

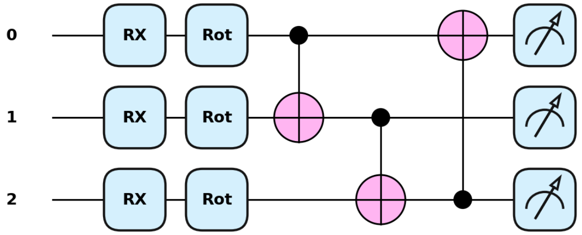


Forecaster + hybrid circuits — printed from the code

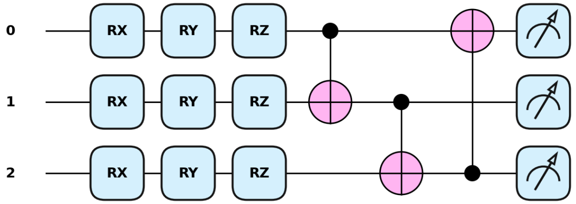
qcpinn — quantum block (n=5, Cascade, L=1)



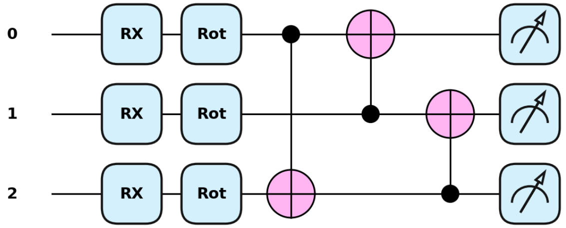
data_reuploading (n=3, L=1)



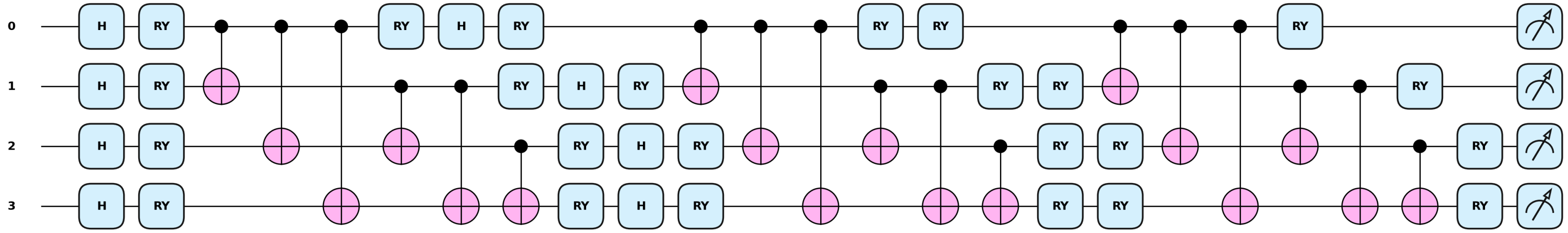
hardware_efficient (n=3, L=1)



strongly_entangling (n=3, L=1, long-range CNOT)

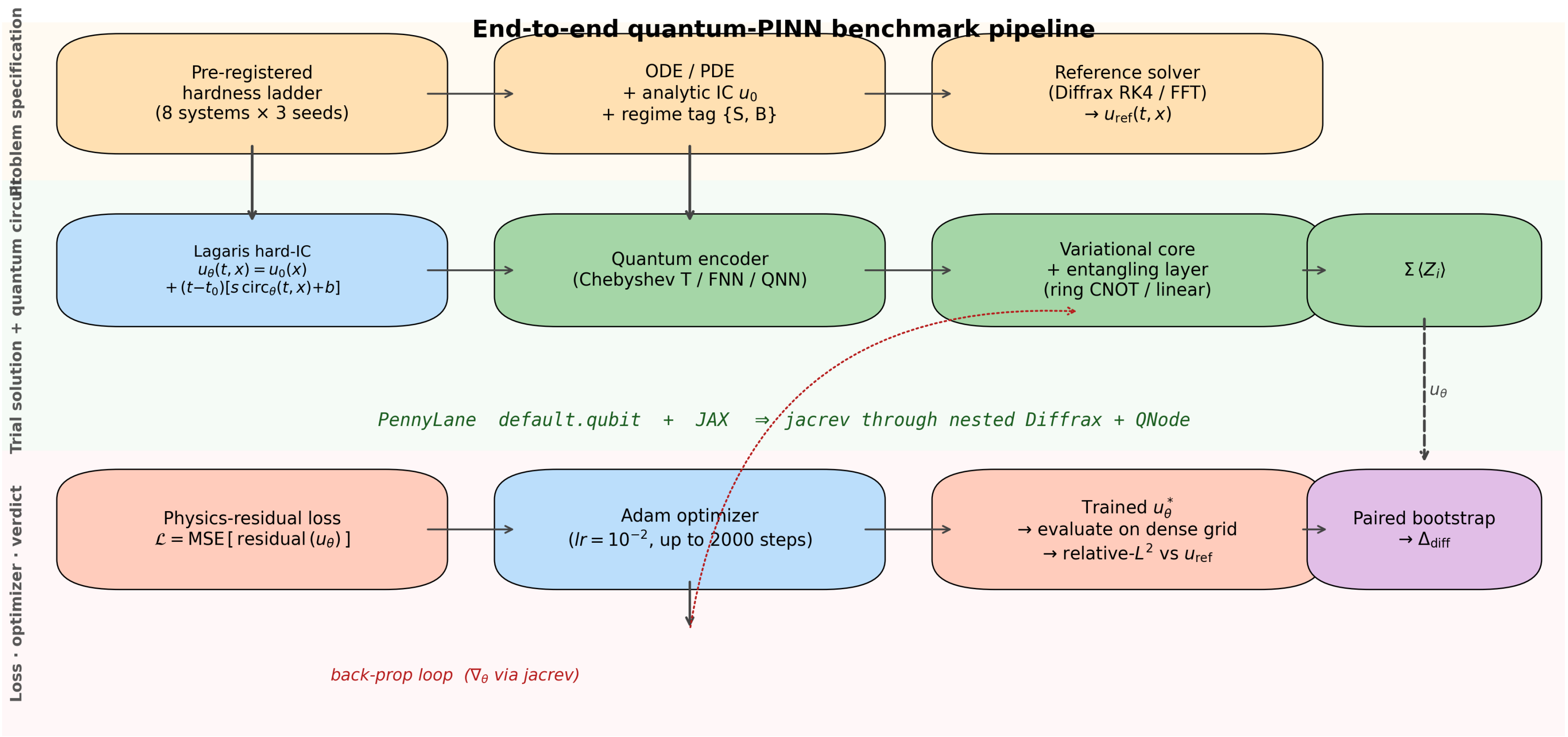


rf_qrc — fixed random-feature reservoir (n=4): double feature map Φ,Φ → frozen V(α) → full 2^n probability readout



Every diagram on these two slides is generated by qml.draw_mpl on the exact circuit objects the training runs execute — zero hand-drawing.

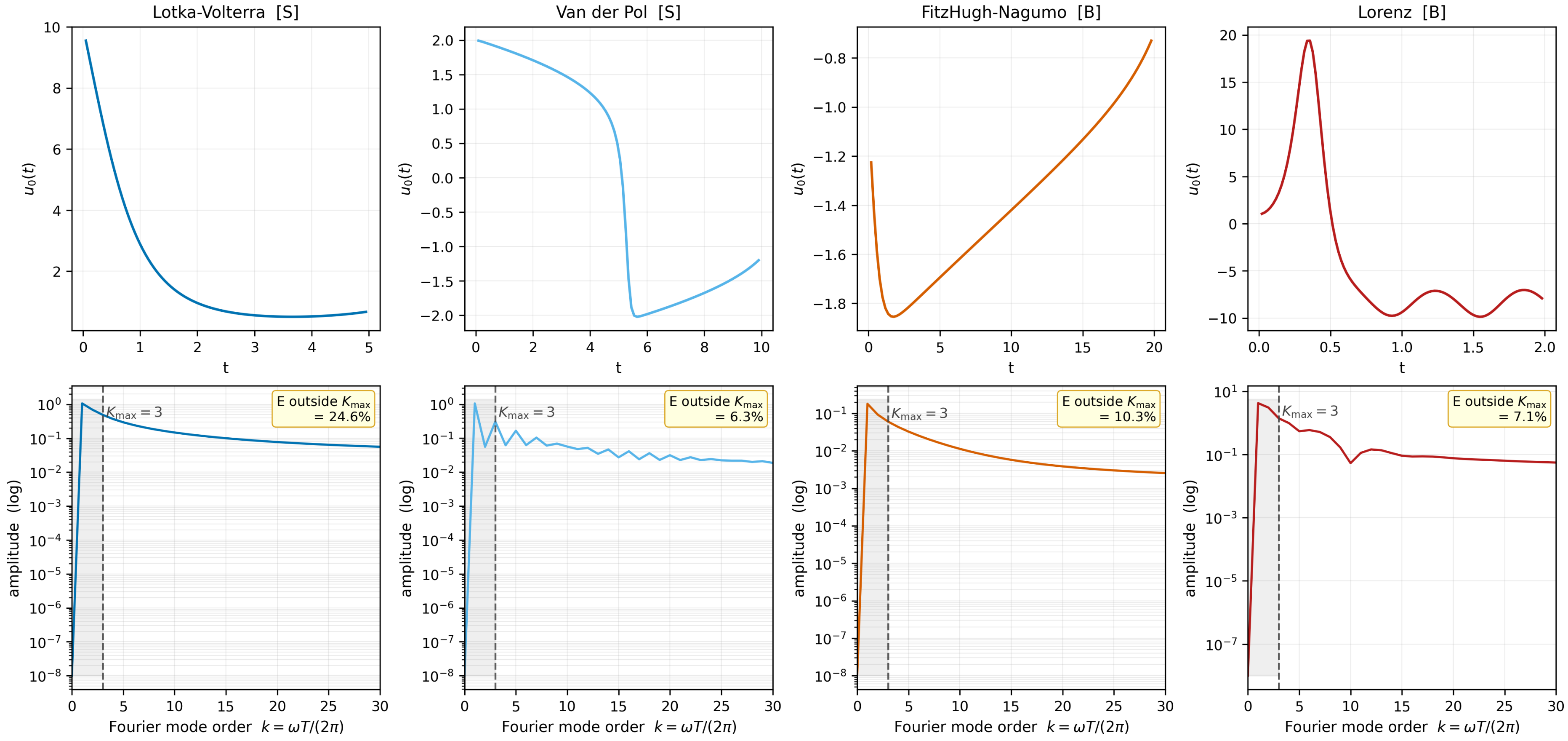
Pre-registered methodology (the Bowles–Schuld checklist)



- **Hypothesis + decision rule locked BEFORE results**
- Matched parameter budgets & training steps
- Symmetric HPO — both sides tuned
- Paired-bootstrap 95% CI (n=10,000)
- Underfit + skyline guards
- 22 amendments — every deviation disclosed
- **verify_paper_integrity.py gates EVERY cited number**

Decision rule (pre-reg §7): CONFIRMED iff 95% CI of Δ_smooth − Δ_broad excludes 0 positively on the solver task.

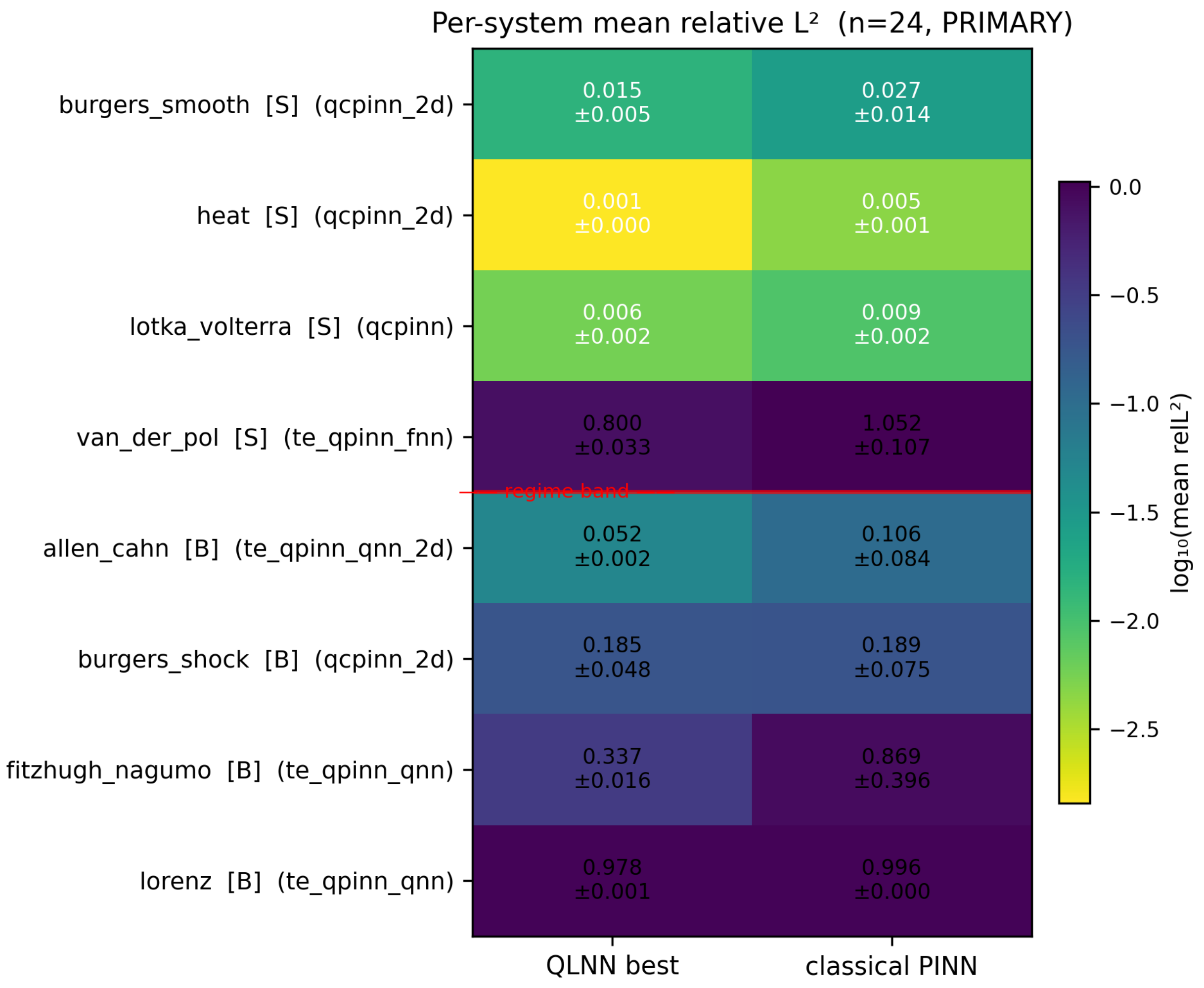
The hardness ladder — 8 systems, 2 regimes



- **SMOOTH / PERIODIC:**
  - Lotka-Volterra · Van der Pol · Heat · Burgers (smooth)
- **BROADBAND / MULTISCALE:**
  - Lorenz · FitzHugh-Nagumo · Allen-Cahn · Burgers (shock)
- Spectra measured, not asserted — PSD of each reference solution
- 3 seeds per cell; Kuramoto (12-D) + KdV running on Anvil now

Regime tags bound in code and asserted against the pre-registration.

Solver task: per-system results (n = 24 cells)



- Quantum wins the easy smooth systems (LV, Heat) — but so does everything; margins are small
- **te_qpinn_qnn is the clear broadband story: best-in-class on FitzHugh-Nagumo AND Allen-Cahn**
- chebyshev family confirmed smooth-only (fails shocks & fronts)
- Lorenz: every model collapses to predict-zero — a shared-failure datapoint

24×

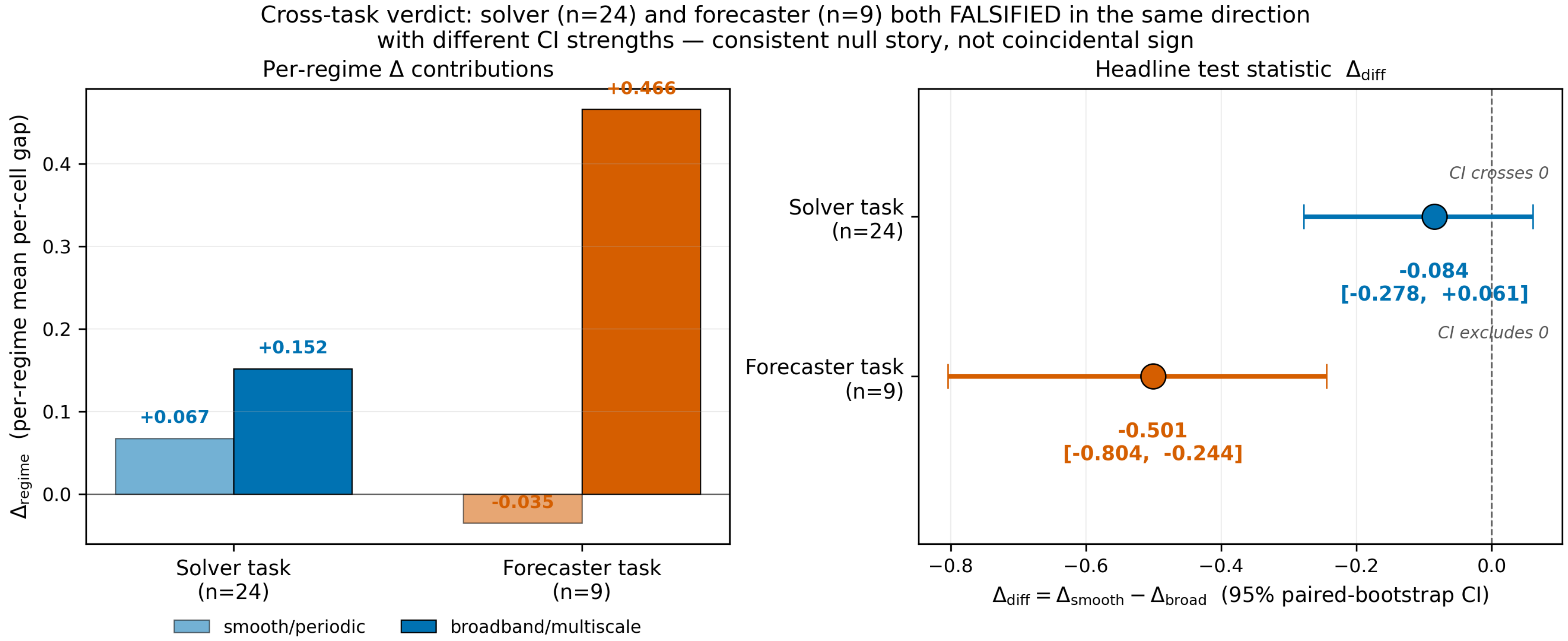
tighter seed std-dev: te_qpinn_qnn vs cPINN on FHN

2.5×

lower mean error on FHN (0 classical params)

Full all-to-all per-cell table in paper §3; every value integrity-gated.

The pre-registered verdicts: H1 is FALSIFIED — on both tasks



−0.084

SOLVER Δdiff (n=24)
CI [−0.278, +0.061]

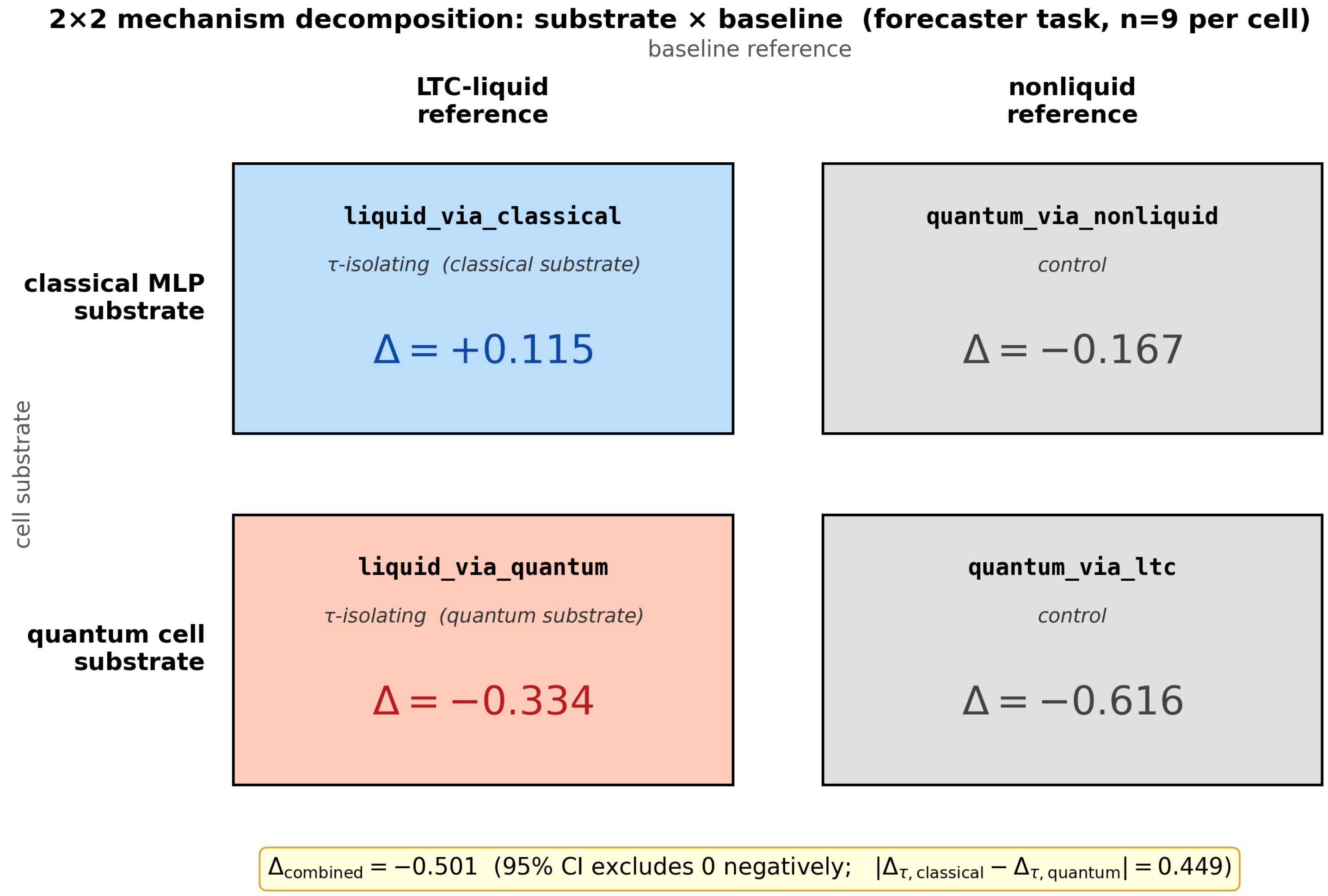
−0.501

FORECASTER Δdiff (n=9)
CI [−0.804, −0.244]

- **Every sensitivity point agrees: expanding n, tuning both sides, adding systems — the predicted smooth-side advantage never appears**
- Forecaster CI excludes zero NEGATIVELY — the trend points the opposite way from the hypothesis

"Published as a rigorous mechanistic null" — the outcome class our pre-registration explicitly planned for.

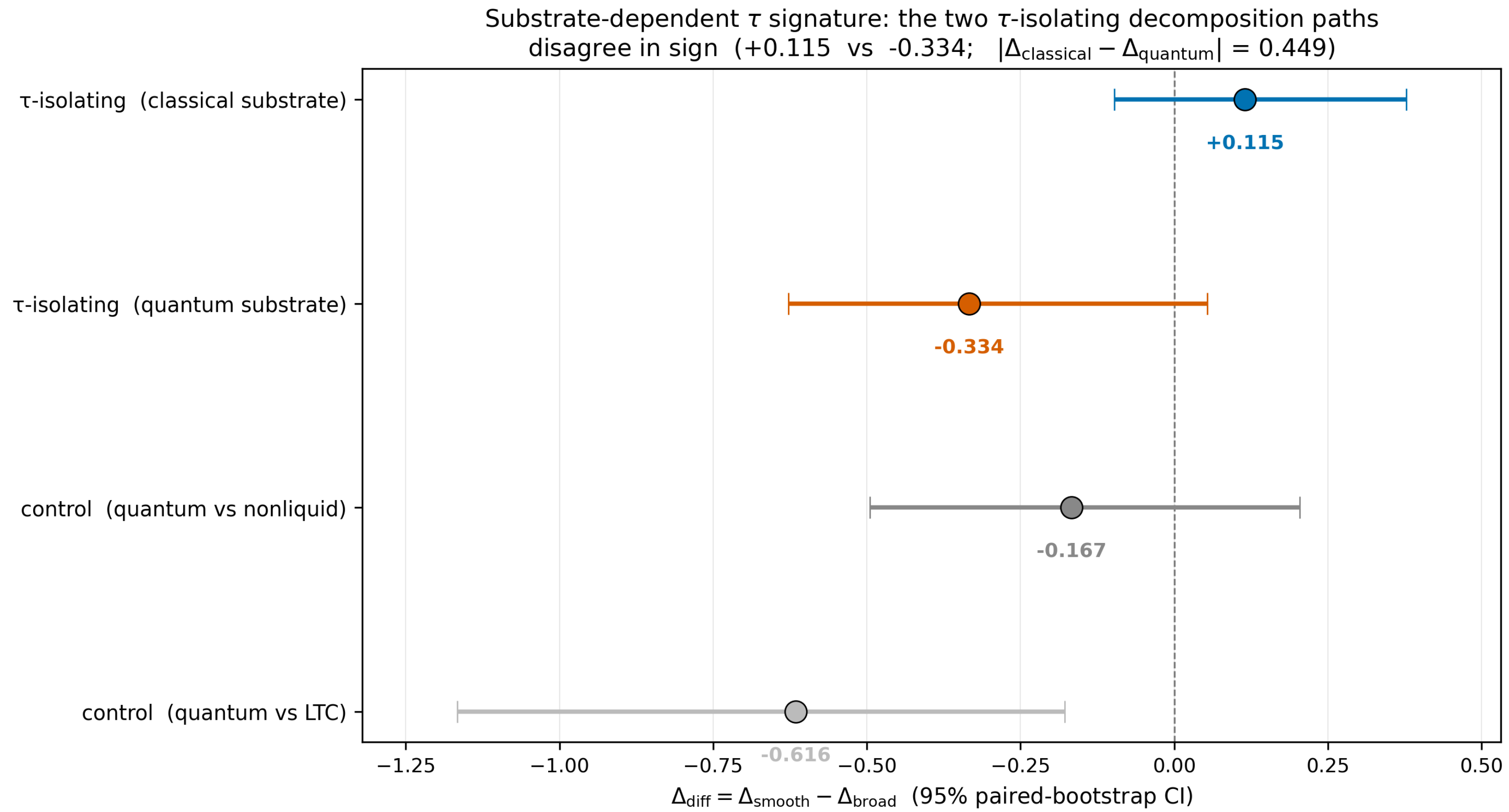
Whose fault is it? The complete fairness 2×2



- **We built all four corners: {classical, quantum} × {liquid τ, non-liquid}**
- Δ_combined = Δ_quantum + Δ_liquid — the algebraic identity holds per-cell EXACTLY, along two independent paths
- **Quantum-isolated contribution: −0.616, CI excludes 0 → the quantum circuit is the source of the underperformance**
- The liquid-τ machinery is NOT the problem

Ablations: classical LTC (P7.10) and non-liquid quantum cell (P7.11) complete the square.

The surprise: liquid τ is substrate-dependent



+0.115

τ added to a CLASSICAL cell → smooth-favoring (matches the original hypothesis!)

−0.334

τ added to the QUANTUM cell → broadband-favoring (sign flips)

- **Identical τ machinery, opposite effect depending on what it wraps — measured along two independent ablation paths**
- The classical-side direction is the ONLY measurement in the whole program matching the Schuld-Fourier prediction

This interaction effect is the seed of the follow-up paper (theory derivation deferred to Tier-4 work).

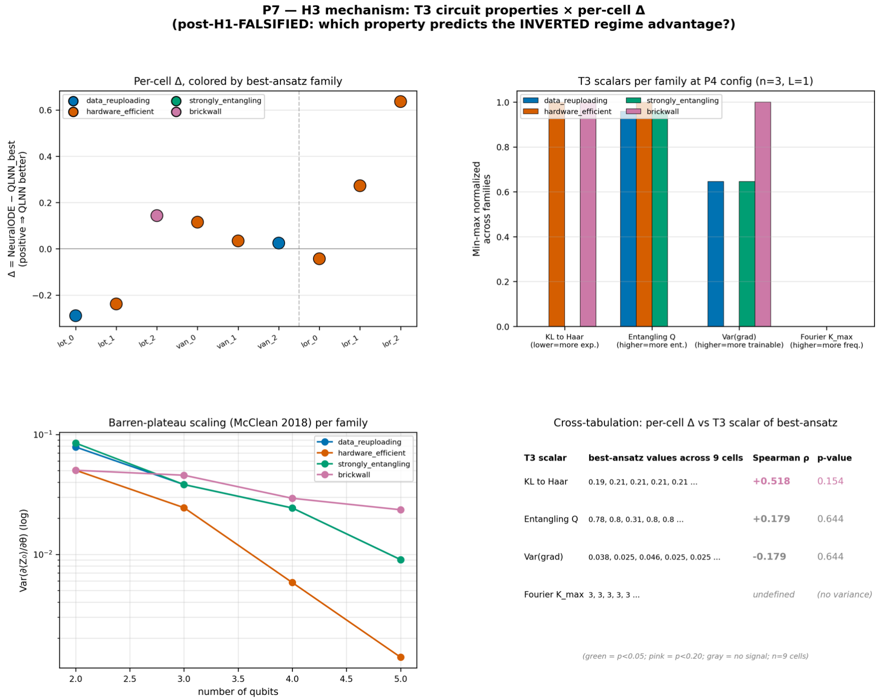
Rigor receipts — why this survives review

36 pp
paper + supplement, built from source

22
pre-reg amendments (A1–A22), all disclosed

37
verified bibliography entries, 22 woven citations

508
tests green + integrity gate exit-0



- 5-reviewer adversarial audit already run internally; every critical finding remediated (4-tier plan)
- **Every number in the paper is machine-checked against committed result JSONs — reviewers can clone & re-verify**
- Mechanism cross-check (left): expressivity (KL-to-Haar) is the only trainability scalar with a sign-stable trend vs per-cell Δ (ρ = +0.518, LOO-robust)
- Open-source: github.com/shawngibford/qlnn — tagged releases v0.1 → v1.2

Where we are & what we need

- **DONE — paper v1.2 (peer-review-hardened), all local experiments, SLURM pipeline**
- IN FLIGHT — Anvil Phase C: 222-cell audit re-run matrix (Kuramoto + KdV completion, uniform budgets, qcpinn attribution, forecaster refresh) — ~1 day wall-clock remaining
- NEXT — Phase D verdict refresh → final paper numbers → arXiv preprint + PRX Quantum submission

Asks for you

- Sign-off on the falsification framing (rigorous null + mechanism finding)
- Venue check: PRX Quantum first, or Quantum / PR Research as the faster path?
- Co-author review pass on the 36-page draft (2 weeks?)